REVIEW · REVIEW

Week 2 recall day

Answers and worked solutions · Day 16 · Sun 8 Nov · pack R2

Part A — Machines short-answer bank

A.1 Past-paper short answers (all with official model points)

A1.

- Many loads, especially induction motors and other inductive loads, draw reactive power as well as real power. Reactive power is energy that oscillates back and forth between the source and the load each cycle. It does no useful work.
- That oscillating energy still needs current. So the supply current, and the apparent power, are larger than the real power alone would need. This can overload generators, transformers and cables, and the extra current causes extra I^2R losses. Supply companies also charge for a low power factor (peak VA).
- Power-factor correction adds a capacitor in parallel with the load. The capacitor generates reactive power, which supplies the load's reactive power locally. So the reactive power drawn from the supply falls.
- The real power and the load's own operation do not change. Only the supply current falls, and the power factor seen by the supply rises towards unity.

OFFICIAL POINTS: loads draw reactive power, which is energy oscillating in the system that can overload generators and transformers and produce extra losses; a parallel capacitor generates reactive power and so reduces the reactive power drawn from the supply.

A2.

Similarities.

- Both can run from a three-phase AC supply. Both have a primary winding (the stator, for an induction machine), a secondary winding (the rotor) and an iron magnetic core.
- Both transfer power from primary to secondary by electromagnetic induction. So both have very similar equivalent circuits: winding resistances, leakage reactances, a magnetising reactance and a core-loss resistance.

Differences.

- In an induction machine the secondary winding rotates and is short-circuited. In a transformer the secondary is stationary and feeds an external load.
- The rotor currents are at slip frequency, not supply frequency. So the rotor frequency changes with speed. In a transformer both sides are at the same frequency.
- In the induction machine equivalent circuit, the rotor resistance is divided by the slip. This variable resistance represents both the rotor copper loss and the mechanical

output power.
- An induction machine has an airgap, so its magnetising current is much larger (about 25–60% of rated current, versus about 10% for a transformer).

OFFICIAL POINTS: both 3ph AC with primary and secondary windings, a magnetic core and similar equivalent circuits; but the IM secondary is rotating and short-circuited and the rotor frequency differs from the stator frequency; in the IM equivalent circuit the rotor resistance is divided by slip.

A3.

- A perfect conductor means the referred rotor resistance is zero.
- There would be no rotor copper loss, so the efficiency would rise.
- Maximum torque occurs at a slip proportional to the rotor resistance. With zero rotor resistance, the torque would rise vertically at synchronous speed. The machine would behave like a synchronous machine: the speed would not drop as load is added.
- The starting torque would be zero. Torque needs power absorbed in the effective rotor resistance, and at standstill that resistance would be zero. So the motor could not start on its own.

OFFICIAL POINTS: zero rotor resistance; the torque–speed slope at synchronous speed becomes vertical (like a synchronous machine); no rotor losses so higher efficiency; zero starting torque. The sheet suggests thinking about the torque curves in Jun23 Q3b(ii).

A4.

- **Real power** is the average of the instantaneous power. It is the net energy flow from source to load per second. It comes from the part of the current that is in phase with the voltage. It does useful work (heat, light, mechanical output) and is absorbed by resistances.
- **Reactive power** comes from the part of the current that is 90° out of phase with the voltage. It is the peak rate of energy that oscillates back and forth between source and load and averages to zero. Inductors absorb it and capacitors generate it. It does no useful work, but it still needs current.
- **Apparent power** is the magnitude of the complex power, whose real part is the real power and whose imaginary part is the reactive power. It is the product of the rms voltage and the total rms current.
- **Practical significance:** apparent power sets the current, so it sets the rating (VA) of generators, transformers and cables. Real power is what the customer pays for as energy. Reactive power is why we use PFC. Real and reactive power are both scalars.

OFFICIAL POINTS: for real and reactive power, any of: phase of the current relative to the voltage, link to instantaneous power, energy flow between source and load, ability to do useful work, which circuit elements absorb them, both scalar. For apparent power: magnitude of complex power, related to the total current.

A5.

- **Copper (winding) loss** is the resistive heating of the primary and secondary windings caused by the current flowing through them. It is modelled by the series winding resistance (the primary and referred secondary resistances combined).
- **Iron (core) loss** is the loss in the magnetic core. It has two parts: hysteresis loss (energy to cycle the iron's magnetisation) and eddy-current loss (currents induced in the iron itself). It is modelled by the core-loss resistance in the shunt branch, across the supply.
- **Variation with output power:** copper loss depends on the square of the current. At constant voltage, it is proportional to the square of the output power. Iron loss depends on the flux, which is set by the supply voltage and frequency. So it is roughly constant and does not vary with output power.
- So at light load the iron loss dominates, and at heavy load the copper loss dominates. Maximum efficiency occurs where the two are equal.

OFFICIAL POINTS: explain copper and iron losses, their physical loss mechanism and their equivalent-circuit parameters; copper loss is proportional to the square of output power, and iron loss does not vary with output power.

A6.

- **Magnetising current.** The stator must set up the rotating magnetic field, even with no load. That current flows in the magnetising reactance. Because the induction machine has an airgap, the reluctance is high, so this current is large, typically 25–60% of rated current. It is almost purely reactive, which is why the no-load power factor is very low.
- **Core loss.** The rotating flux produces hysteresis and eddy-current losses in the stator iron. The stator must draw a small in-phase current to supply them. This is the current in the core-loss resistance.
- **Mechanical losses.** Even with no shaft load, the rotor must overcome friction in the bearings and windage (air drag, including the cooling fan). So the rotor runs slightly below synchronous speed, the slip is small but not zero, and a small rotor current flows to supply that power.
- (A small stator copper loss also occurs because these currents flow through the stator resistance.)

OFFICIAL POINTS: magnetising current and core-loss resistance, with their equivalent-circuit parameters and physical reasons; mechanical losses (friction and windage) need some power even at no load.

A7.

Process.

- For a given voltage and frequency, the back-emf equation fixes the product of the turns and the peak flux. So the number of turns sets the peak flux, and hence the peak flux density, in the core.
- Choose the number of turns so that the peak flux density is just below the saturation value of the core material.

Too few turns.

- The peak flux density needed is too high. The core saturates.
- In saturation the magnetising current rises very sharply, so the input current is high and distorted. The winding may overheat.

Too many turns.

- The flux density is lower than necessary, so the core is under-used.
- More turns of wire mean a longer winding, so a higher winding resistance (and if the wire must be thinner to fit the window, higher still). The copper loss rises and the efficiency falls. The winding may overheat.

`OFFICIAL POINTS:` the turns change the peak core flux (back-emf equation) and are chosen so the flux density is below saturation; too few turns gives too high a flux density, saturation, high input current and possible overheating; too many gives low flux density, higher winding resistance, higher copper loss and lower efficiency, and possible overheating.

A8.

Ideal transformer.

- The output voltage does not change. It is set only by the input voltage and the turns ratio, and the load current does not affect it.
- The output current halves. The voltage is fixed and the resistance has doubled (Ohm's law).

Real transformer.

- The output voltage rises slightly. A real transformer has series winding resistance and leakage reactance. When the load current falls, the voltage drop across them falls, so more of the supply voltage reaches the load.
- The output current still falls, but to slightly more than half its original value, because the output voltage is now slightly higher.

> **OFFICIAL POINTS:** the same four points, about 1 mark each. (This was also in R1 Part A.)

A9.

- **Lagging.**
- The transformer's series impedance is mainly leakage reactance. So the load current lags the primary voltage that drives it through that reactance.
- The load is purely resistive, so the load (secondary) voltage is in phase with the load current.
- Therefore the secondary voltage lags the primary voltage.

Official answer: lagging, because the series reactance makes the current lag the primary voltage, and the load voltage is in phase with the current for a resistive load.

A10.

- The rated speed is just below a synchronous speed. $60 \times 50/560 = 5.357$, so round down to $p = 5$ pole-pairs: **10 poles**, synchronous speed **600 rpm**.
- **Induction machine**, because the rated speed (560 rpm) is below the synchronous speed (600 rpm). A synchronous machine runs exactly at synchronous speed.
- Ideal efficiency (rotor copper loss only) $= n/n_S = 560/600 =$ **93.33%**.

This part is half words, half numbers. In words: the rotor must run slower than the field to have slip, and the fraction of airgap power lost in the rotor equals the slip.

> **OFFICIAL ANSWERS:** 10 poles; induction, as the rated speed is less than synchronous speed; 93.33%.

A11.

- **DC machine:** halve the terminal voltage. The no-load speed is proportional to voltage. The torque–speed slope is unchanged, so the curve slides left.
- **Synchronous machine:** halve both the frequency and the voltage. Halving the frequency halves the synchronous speed. Halving the voltage too keeps the volts-per-hertz ratio constant, so the peak flux stays at its rated value and the machine does not saturate.
- **Induction machine:** halve both the frequency and the voltage, for the same reason. The torque–speed curve keeps its shape and slides left, and the maximum torque is unchanged.
- "Safe" here means: never halve only the frequency on an AC machine. The flux would double, the core would saturate and the magnetising current would become very large.

> **OFFICIAL ANSWERS:** synchronous: halve both V and f; DC: halve V; induction: halve both V and f.

A12.

- On open circuit the only current is in the shunt branch (core-loss resistance in parallel with magnetising reactance). That branch is linear in the simplified model.
- **Current: ×0.5.** Current is proportional to voltage through a fixed impedance.
- **Real power: ×0.25.** Real power is the voltage squared over the core-loss resistance.
- **Reactive power: ×0.25.** Reactive power is the voltage squared over the magnetising reactance.
- **Power factor: ×1.** Both powers scale by the same factor, so their ratio, and the impedance angle, is unchanged.

`OFFICIAL ANSWERS:` ×0.5, ×0.25, ×0.25, ×1. (A real core would saturate less at lower voltage, so the magnetising reactance would rise a little, but the linear model gives these factors.)

A.2 Likely new short answers on Ch4–8

N1.

- Balanced three-phase currents in the stator winding produce a magnetic field of constant magnitude that rotates at synchronous speed.
- If the rotor turns slower than this field, the rotor conductors see a changing field. This induces voltages in them, proportional to the slip speed. The rotor winding is short-circuited, so these voltages drive rotor currents.
- The rotor currents interact with the stator field to produce a force on each conductor, and so a torque. The field effectively drags the rotor around.
- At synchronous speed the rotor moves with the field, sees no change in flux, has no induced voltage and no current, so the torque is zero. Some slip is always needed to produce torque.

N2.

- **DC resistance test:** measure the DC resistance of the stator winding. This gives the stator resistance per phase. (Line-to-line on a star machine gives twice the phase value.)
- **No-load test:** run the motor unloaded at rated voltage and record voltage, current and power. The slip is nearly zero, so the effective rotor resistance is nearly infinite and the rotor branch carries almost no current. The current is small, so the series drops can be ignored. The test gives the core-loss resistance and magnetising reactance (parallel model).
- **Locked-rotor test:** stop the rotor turning (slip = 1) and apply a reduced voltage that gives rated current. The voltage is low, so the shunt branch can be ignored. The test gives the total series resistance and the total leakage reactance (series model).

- Subtract the stator resistance from the total series resistance to get the referred rotor resistance. The leakage reactance is usually split equally between stator and rotor.

N3.

- From the speed and the synchronous speed, find the slip. This fixes the effective rotor resistance, so every element in the per-phase circuit is now known.
- Apply the per-phase supply voltage and solve the circuit as an ordinary AC circuit: rotor branch current, magnetising and core-loss currents, total input current, input power factor and input power (three times the per-phase value).
- The power absorbed by the effective rotor resistance is the airgap power. Dividing it by the synchronous angular speed gives the torque.
- The mechanical output power is the torque times the actual rotor speed (equivalently, one minus slip times the airgap power). Comparing it with the input power gives the efficiency.
- The model assumes balanced three-phase steady state. It does not predict start-up transients.

N4.

- **Squirrel-cage rotor:** solid (usually cast aluminium) bars shorted by end rings. The rotor resistance is fixed. It is cheap, robust and the most common type.
- **Wound rotor:** a three-phase winding like the stator, brought out through slip rings and brushes. External three-phase resistance can be added, so the effective rotor resistance can be changed.
- Adding rotor resistance increases the slip at which maximum torque occurs, but does not change the value of the maximum torque.
- So at starting it can reduce the starting current and increase the starting torque (even giving maximum torque at standstill). It is then removed to restore efficiency. It also gives limited, inefficient speed control for fan loads.

N5.

- An inverter supplies the stator with three-phase voltage of adjustable magnitude and frequency. Changing the frequency changes the synchronous speed, and so the operating speed.
- By the back-emf equation, the peak flux is proportional to the voltage divided by the frequency (neglecting stator resistance). If the frequency is reduced at constant voltage, the flux rises and the core saturates.
- So the voltage is varied in proportion to the frequency, keeping the volts-per-hertz ratio at its rated value. The flux stays constant.
- Result: the torque–speed curve keeps its shape and just slides along the speed axis. The maximum torque stays the same, the slip stays small (so efficiency stays high), and maximum torque is even available at standstill.

- Limits: speed control is not exact (it still depends on slip) and dynamic performance is limited. It suits fans, pumps and conveyors.

N6.

- The stator is a standard three-phase AC stator. Balanced three-phase currents in it produce a magnetic field rotating at synchronous speed.
- The rotor carries DC field current (fed through slip rings) or permanent magnets. This gives a rotor field fixed relative to the rotor, so it rotates with the rotor.
- The two fields try to align, like two discs joined by a spring. The rotor turns at exactly synchronous speed, with no slip. The angle between the fields (the rotor angle) sets the torque and power.
- As a generator, the rotor field leads and drags the stator field. As a motor, the stator field leads and drags the rotor. If the torque needs a rotor angle beyond 90°, the machine pulls out of synchronism, which causes large currents and oscillating torques.
- It is not self-starting from fixed-frequency mains, because the stator field rotates too fast for the rotor to lock on from standstill.

N7.

- The generator must produce 50 Hz, so the number of poles is set by the turbine speed. Hydro turbines are slow (about 50–300 rpm), so they need many poles (20 or more). Steam and gas turbines run at 1500 or 3000 rpm, so they need only 4 or 2 poles.
- **Salient poles** project out from the rotor body, so it is easy to fit many of them around a large-diameter rotor. That suits slow, many-pole hydro generators.
- **Round rotors** embed the field winding in slots in a solid cylindrical rotor. This is mechanically strong enough to resist the huge centrifugal forces at high speed, and it keeps the diameter small. That suits 2- and 4-pole thermal and nuclear generators.

N8.

- The stator winding has a maximum current, so the machine has an apparent power (VA) rating at rated voltage.
- The real and reactive power together must keep the apparent power within that rating. So the more real power the machine delivers, the less reactive power it can generate or absorb.
- At zero real power it can supply or absorb reactive power up to its full VA rating. At full rated real power it can provide no reactive power at all.
- (Other limits: field heating limits how far it can be over-excited, and stability limits how far it can be under-excited, because the rotor angle must stay well below 90°.) A synchronous compensator is a machine run with no mechanical load, purely to control reactive power.

N9.

- Most industrial loads are induction motors. Induction motors need magnetising current to set up their airgap field. This current is reactive (inductive), so the plant draws lagging reactive power. The effect is worst at light load.
- A synchronous motor runs at constant speed, and its reactive power is controlled by its field current. When over-excited it generates reactive power, so it behaves like a PFC capacitor while it also drives a mechanical load.
- This raises the plant's overall power factor and reduces the supply current and the peak-VA (power-factor) charges from the supply company.
- Synchronous motors cost several times more than induction motors, but the lower tariff can justify the cost.

N10.

- The power absorbed by the effective rotor resistance is the airgap power, the power crossing from stator to rotor.
- One part is the actual referred rotor resistance. The power it absorbs is real heat in the rotor winding (rotor copper loss), and it equals the slip times the airgap power.
- The other part is an "imaginary" resistance (the effective resistance minus the actual one). The power it absorbs is the electrical power converted to mechanical output power, which equals one minus slip times the airgap power.
- So, counting only rotor copper loss, the ideal (maximum possible) efficiency is one minus the slip, for example 95% at a slip of 0.05.
- The actual efficiency is lower, because stator copper loss, iron loss and friction and windage losses also occur.

N11.

- Each quantity is expressed as a fraction of a chosen base value (normally the machine's VA rating and its rated voltage). So you see at once how large it is relative to normal. A voltage of 5 pu is obviously wrong, while 139 kV may or may not be.
- In per-unit, line and phase voltage have the same value, and per-phase and total power have the same value. So the factors of $\sqrt{3}$ and 3 disappear, which removes a major source of errors.
- With a transformer, the per-unit values are the same on both sides when the bases follow the turns ratio, so referring is automatic.
- The choice of base does not change the final answer in real units. Machine parameters in per-unit (such as synchronous reactance) fall in narrow typical ranges, so values from machines of very different sizes can be compared.

N12.

- **Open-circuit test:** drive the machine at synchronous speed with another motor, with the stator open-circuited. Measure the open-circuit stator voltage as the field current is varied. It rises linearly at first, then bends over as the iron saturates.
- **Short-circuit test:** drive it at synchronous speed with the stator short-circuited. Measure the stator current as the field current is varied. The result is a straight line.
- In the short-circuit test, the field produced by the stator current is nearly equal and opposite to the rotor field. So the net flux is small and the iron never saturates. On open circuit there is no stator current, so the full rotor field is present.
- At a given field current, the synchronous impedance is the open-circuit phase voltage divided by the short-circuit current. The synchronous reactance follows after removing the stator resistance. At high field currents saturation makes the synchronous reactance fall.

Part B — Mixed recall

B1.

$n_S = 60f/p = 60 \times 50/3 = \mathbf{1000\ rpm}$. $s = (1000 - 960)/1000 = \mathbf{0.04}$.
$f_2 = sf = 0.04 \times 50 = \mathbf{2\ Hz}$. Ideal efficiency $= 1 - s = \mathbf{96\%}$. This is the lecture 7.6.2 machine.

B2.

With $W_4 = e^{-j\pi/2} = -j$, $X(k) = \sum_n x(n)(-j)^{kn}$:

- $X(0) = 1 + 2 + 0 + 1 = 4$
- $X(1) = 1 + 2(-j) + 0 + 1(j) = 1 - j$
- $X(2) = 1 - 2 + 0 - 1 = -2$
- $X(3) = 1 + 2(j) + 0 + 1(-j) = 1 + j$

$X(k) = \{\mathbf{4,\ 1 - j,\ -2,\ 1 + j}\}$. Check: x is real, so $X(3) = X^*(1)$. And $\frac{1}{4}\sum X(k) = 1 = x(0)$.

B3.

$g_m = I_C/V_T = 1\ \text{mA}/26\ \text{mV} = \mathbf{38.46\ mA/V}$. $I_E = I_C(\beta + 1)/\beta = 1.01$ mA, so $r_e = 26\ \text{mV}/I_E = \mathbf{25.74\ \Omega}$ (26 Ω if you take $I_E \approx I_C$).
$r_\pi = \beta/g_m = (\beta + 1)r_e = \mathbf{2.6\ k\Omega}$.

B4.

$\tau = kI = 0.15 \times 5 = \mathbf{0.75\ Nm}$. $E = V - IR = 24 - 5 \times 0.4 = 22$ V. $\omega = E/k = 146.7$ rad/s, so $n = 146.7 \times 60/2\pi = \mathbf{1401\ rpm}$. $P_{in} = VI = 120$ W, $P_{out} = \tau\omega = 110$ W, $\eta = \mathbf{91.67\%}$ (equal to E/V). Trap: k uses rad/s, so convert to rpm only at the end. The no-load speed is $V/k = 160$ rad/s (1528 rpm), and the stall current is $V/R = 60$ A.

B5.

Shortcut: linear convolution first, $\{1, 1, 2, 3, 1, 0, 0\}$. Then fold the samples at $n \geq 4$ back by 4 and add: $y(0) = 1 + 1 = 2$, and $y(1), y(2), y(3)$ stay $1, 2, 3$.

$$y(n) = x_1(n) \circledast_4 x_2(n) = \{\underline{2}, 1, 2, 3\}$$

Check: $\sum y = 8 = (\sum x_1)(\sum x_2) = 4 \times 2$.

B6.

First step: DC analysis to get I_E, because r_e depends on the bias point. Check whether the approximate method is allowed: $\beta R_E = 100$ kΩ $\geq 10R_2 = 82$ kΩ, so yes.

- $V_B = 12 \times 8.2/(39 + 8.2) = 2.085$ V, $V_E = 1.385$ V, $I_E = 1.385$ mA, so $r_e = 26\ \text{mV}/1.385\ \text{mA} = \mathbf{18.78\ \Omega}$.
- $R' = R_1 \parallel R_2 = 6.775$ kΩ. $\beta r_e = 1.878$ kΩ. $Z_i = R' \parallel \beta r_e = \mathbf{1.470\ k\Omega}$.
- $Z_o = R_C = \mathbf{3.3\ k\Omega}$.
- $A_v = -R_C/r_e = \mathbf{-175.8}$ (the minus sign means 180° phase shift).

(The exact Thevenin bias gives $I_E = 1.298$ mA and $r_e = 20.04\ \Omega$. Use whichever your course question asks for, and say which.)

B7.

1. **Per-phase voltage:** $V_1 = 415/\sqrt{3} = 239.6$ V (star).
2. **Poles and synchronous speed** from the rated speed (n_S just above it), then **slip** $s = (n_S - n)/n_S$ and the **effective rotor resistance** R_2'/s.
3. **Rotor branch current:** $I_2' = V_1/(R_1 + R_2'/s + jX)$. Add $V_1/R_C + V_1/jX_M$ for I_1 if the shunt branch is included.
4. **Torque:** $\tau = 3I_2'^2(R_2'/s)/\omega_S$, with $\omega_S = 2\pi n_S/60$.

Traps: use ω_S, not the rotor speed, in the torque equation. Don't forget the factor 3. Keep the slip to 4 significant figures, because R_2'/s is sensitive to it.

B8.

You need $N \geq L + M - 1 = 7$. The next power of 2 is $\mathbf{N = 8}$. Zero-pad both sequences to 8, multiply the DFTs, and take the IDFT. With $N = 6$ the result is time-aliased: the last linear-convolution sample (at $n = 6$) wraps onto $n = 0$, so the first $7 - 6 = 1$ sample is corrupted.

B9.

$Z_b = \beta(r_e + R_E) = 100 \times (18.78 + 1000) = 101.9$ kΩ.
$Z_i = R' \parallel Z_b = 6.775\ \text{k} \parallel 101.9\ \text{k} = \mathbf{6.353\ k\Omega}$. $A_v = -R_C/(r_e + R_E) = \mathbf{-3.239}$ ($\approx -R_C/R_E = -3.3$). β has dropped out, and r_e is small next to R_E. So the gain is set by two resistors and hardly changes with temperature or transistor spread. The price is a much lower gain (compare Prac 3, bypassed versus unbypassed).

B10.

Maximum torque occurs at maximum airgap power, when R_2'/s equals the magnitude of the rest of the series impedance:

$$|R_1 + jX| = |0.5 + j3| = 3.041\ \Omega, \qquad s_{\tau\max} = 0.6/3.041 = \mathbf{0.1973}$$

At standstill $s = 1$, so $R_2' + R_{2EXT}' = 3.041\ \Omega$, giving $R_{2EXT}' = \mathbf{2.441\ \Omega}$. The maximum torque value does not change. Only its slip moves (Jun23 Q3b(i) uses the same method).

B11.

- $F_s \geq 2B = \mathbf{8\ kHz}$.
- Resolution $= F_s/N \leq 25$ Hz, so $N \geq 320$. Round up to a power of 2: $\mathbf{N = 512}$.
- $T_0 = N/F_s = 512/8000 = \mathbf{64\ ms}$.

B12.

$Z_b = \beta(r_e + R_E) = 100 \times 2210 = 221$ kΩ. $Z_i = R_B \parallel Z_b = \mathbf{110.2\ k\Omega}$.
$Z_o = R_E \parallel r_e = \mathbf{9.955\ \Omega}$. $A_v = R_E/(R_E + r_e) = \mathbf{0.9955}$, in phase. That is why it is used as a buffer: high Z_i, low Z_o, gain just under 1.

B13.

Per phase: $V_1 = 100/\sqrt{3} = 57.74$ V, $P_1 = 600/3 = 200$ W, $S_1 = V_1 I_1 = 577.4$ VA, $Q_1 = \sqrt{S_1^2 - P_1^2} = 541.6$ VAr. $R = P_1/I^2 = \mathbf{2.0\ \Omega}$, so $R_2' = 2.0 - 1.2 = \mathbf{0.8\ \Omega}$.

$X = Q_1/I^2 = \mathbf{5.416\ \Omega}$ (split $X_1 = X_2' = 2.708\ \Omega$ if needed). Series model, because at $s = 1$ the shunt branch is negligible. Divide by 3 and by $\sqrt{3}$ first.

B14.

The bin spacing is $F_s/N = 31.25$ Hz. $k = 40$ gives $40 \times 31.25 = \mathbf{1250\ Hz}$.
$k = 216 = 256 - 40$ is the mirror image ($X(N-k) = X^*(k)$ for a real signal), not a second tone. Zero-padding to 1024 points makes the bin spacing 7.8125 Hz, which samples the DTFT more finely. But the true resolution, the ability to separate two tones, is still $1/T_0 = F_s/L = 31.25$ Hz, set by the 256 real samples. **No.**

B15.

	Z_i	Z_o	$\lvert A_v \rvert$	Phase
CE (bypassed)	medium ($R' \parallel \beta r_e$, kΩ)	R_C (kΩ)	high (R_C/r_e)	180°
Emitter follower	high ($R_B \parallel \beta(r_e + R_E)$)	very low ($\approx r_e$)	just under 1	0°
Common base	very low ($\approx r_e$, a few Ω to 50 Ω)	R_C	high (R_C/r_e)	0°

The CB current gain is about 1 (α), and the EF current gain is large.

B16.

$I_B = S_B/(\sqrt{3}V_B) = 100\ \text{M}/(\sqrt{3} \times 11\ \text{k}) = \mathbf{5249\ A}$.
$Z_B = V_B^2/S_B = (11\ \text{k})^2/100\ \text{M} = \mathbf{1.21\ \Omega}$ (the same as $(V_B/\sqrt{3})/I_B$).
$X = 1.5/1.21 = \mathbf{1.240\ pu}$.

B17.

$(0.5)^n u(n) \leftrightarrow \dfrac{1}{1 - 0.5z^{-1}}$ for $|z| > 0.5$. $-2^n u(-n-1) \leftrightarrow \dfrac{1}{1 - 2z^{-1}}$ for $|z| < 2$.

$$X(z) = \frac{1}{1 - 0.5z^{-1}} + \frac{1}{1 - 2z^{-1}}, \qquad \text{ROC: } \mathbf{0.5 < |z| < 2}$$

The ROC is the intersection, a ring. It contains the unit circle, so **yes**, the DTFT exists (the signal is absolutely summable).

B18.

Two-port model: the output is a voltage divider between Z_o and R_L, and the input is a divider between R_s and Z_i. $A_{vL} = A_{vNL}\dfrac{R_L}{R_L + Z_o} = -200 \times \dfrac{4}{8} = \mathbf{-100}$.
$A_{vs} = A_{vL}\dfrac{Z_i}{Z_i + R_s} = -100 \times \dfrac{2}{3} = \mathbf{-66.67}$. The loaded gain is always less than the no-load gain.

B19.

$I = 1\angle -36.87^\circ$ pu. $E_0 = V + jIX_S = 1 + j1.2 \times 1\angle -36.87^\circ = \mathbf{1.970\angle 29.17^\circ\ pu}$, so $\delta = \mathbf{29.17^\circ}$. Check: $P = (VE_0/X_S)\sin\delta = 0.8$ pu. ✓
$P_{\max} = VE_0/X_S = 1 \times 1.970/1.2 = \mathbf{1.641\ pu}$ at $\delta = 90^\circ$. $E_0 > V$, so it is over-excited, which is what generating Q (lagging pf) needs. Compare Jun24 Q4b, where 0.9 leading gives $\delta = 82.42^\circ$ (under-excited).

B20.

- $|z| > 1.2$: **causal, unstable** (the ROC does not contain the unit circle).
- $0.5 < |z| < 1.2$: **non-causal (two-sided), stable** (it contains the unit circle).
- $|z| < 0.5$: **anticausal, unstable**.

No ROC gives a system that is both causal and stable, because a pole lies outside the unit circle. Rule: stable means the ROC includes $|z| = 1$, and causal means the ROC is outside the outermost pole.

B21.

Stage 1 is loaded by Z_{i2}: $A_{v1} = -150 \times \dfrac{1.5}{1.5 + 3} = -50$. Stage 2 is loaded by R_L:
$A_{v2} = -120 \times \dfrac{10}{10 + 2.2} = -98.36$. $A_{vT} = A_{v1}A_{v2} = \mathbf{+4918}$ (two inversions, so in phase). The mistake is multiplying the no-load gains ($18\,000$), which overstates the gain by a factor of 3.7. Each stage's load is the next stage's input impedance (Week 5 lecture example, Prac 4).

B22.

$n_S = 1500$ rpm. $s = (1500 - 1530)/1500 = \mathbf{-0.02}$. $R_2'/s = 0.4/(-0.02) = \mathbf{-20\ \Omega}$.
$f_2 = sf = \mathbf{-1\ Hz}$ (the negative sign means reverse phase sequence in the rotor). $n > n_S$, so it is a **generator**. A negative resistance absorbs negative power, so

$P_{AG} = 3I_2'^2 R_2'/s < 0$ and $\tau = P_{AG}/\omega_S < 0$. Mechanical power flows in and electrical power flows out (Jun24 Q3c: $P_{in} = -1451$ W, $\tau = -14.15$ Nm).

B23.

First step: check that it is proper in z^{-1} (numerator degree 0 < denominator degree 2, so yes). Then do partial fractions in z^{-1}:

$$X(z) = \frac{A}{1 - 0.5z^{-1}} + \frac{B}{1 - 0.25z^{-1}}, \quad A = \frac{1}{1 - 0.25/0.5} = 2, \quad B = \frac{1}{1 - 0.5/0.25} = -1$$

The ROC is outside both poles, so both terms are causal:

$$x(n) = [2(0.5)^n - (0.25)^n]u(n)$$

Check by long division: $x(0) = 1$, $x(1) = 0.75$, $x(2) = 0.4375$. Both methods agree.

B24.

a. Shockley: $I_D = I_{DSS}(1 - V_{GS}/V_P)^2 = 10(1 - 0.375)^2 = \mathbf{3.906\ mA}$.
$V_{GS} = V_P\left(1 - \sqrt{I_D/I_{DSS}}\right) = -4(1 - 0.5) = \mathbf{-2.0\ V}$. This is the shorthand point: $I_{DSS}/4$ at $V_P/2$.

b. $I_D = k(V_{GS} - V_T)^2 = 0.3 \times 3^2 = \mathbf{2.7\ mA}$. Saturation needs $V_{DS} \geq V_{DS,sat} = V_{GS} - V_T = \mathbf{3\ V}$. Shockley's equation does **not** apply to the E-MOSFET, and $I_D = 0$ for $V_{GS} < V_T$.